Locally Equivalent Weights for Multilevel Regression and Poststratification

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Running example: Same-Sex Marriage Survey

Analysis of US state-level support for same-sex marriage[1].

- **Survey population:** Five consistently-coded national polls from 2004 ($N_S = 6,341$)
- **Response:** $y = 1$ encodes support for same-sex marriage, $y = 0$ opposition or no opinion.
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We'll analyze this data with two methods: **calibration weighting** (CW) and **MrP** (multilevel regression and poststratification).

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Two methods

Survey: Observe $(\mathbf{x}_i, y_i)$ for $i = 1, \ldots, N_S$
Let $Y_{\mathcal{S}} = \{y_1, \ldots, y_{N_S}\}$.

Target: Observe $\mathbf{x}_j$ for $j = 1, \ldots, N_T$.
We want $\mu := \frac{1}{N_T} \sum_{j=1}^{N_T} y_j$, but don't observe y_j.

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- Frequentist variability
- Regressor balance / External consistency
- Subgroup contribution

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When using OLS, $\hat{y}_j(Y_{\mathcal{S}})$ is linear [Gel07b; BFH21; CZ23], so MrP is a CW estimator.

But currently, for (ubiquitous) MrP models like hierarchical logistic regression, no such equivalent weights are available.[Gel07a]

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CW used raking on the following coarsened regressors: (`survey::calibrate` Lumley [Lum24]):

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Raw survey mean: $\frac{1}{N_S}\sum_{i=1}^{N_S} y_i = 0.333$

Raking: $\hat{\mu}^{\text{CW}}(Y_{\mathcal{S}}) = 0.345$

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How are MrP and raking are using the data differently?

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Linear and nonlinear models

Calibration weighting estimators:

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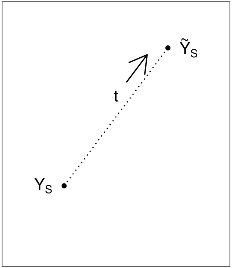
Path through response space



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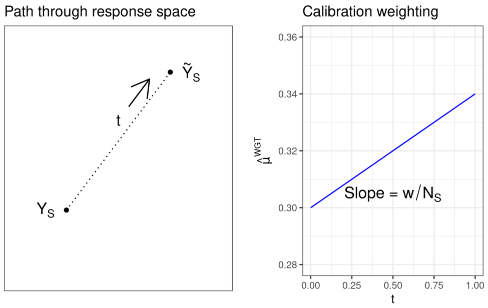

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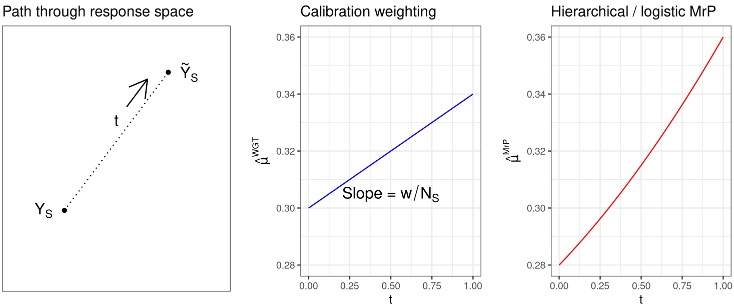

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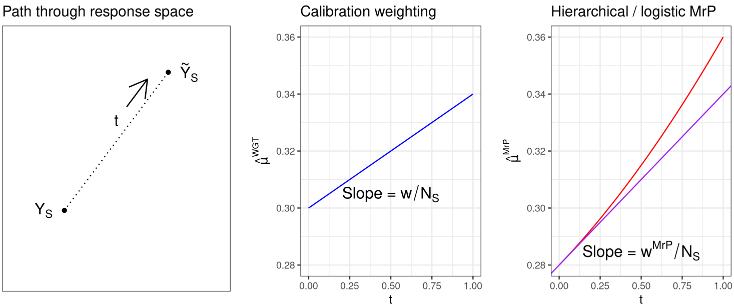

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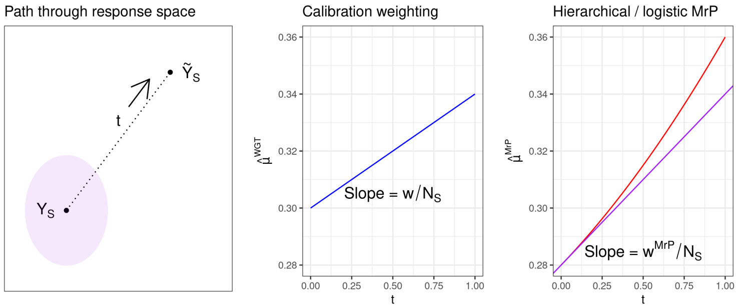

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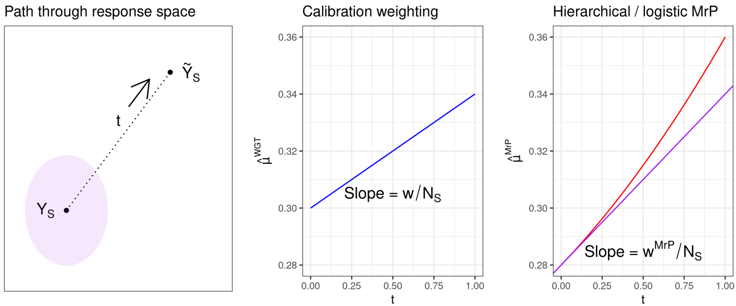

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The weight w_i^{MrP} measures the effect of local perturbations to y_i on the estimator $\hat{\mu}^{\text{MrP}}$.

Our main contribution is to rigorously connect this idea to CW diagnostics!

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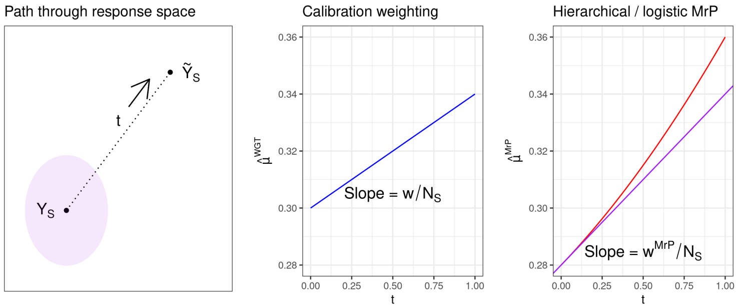

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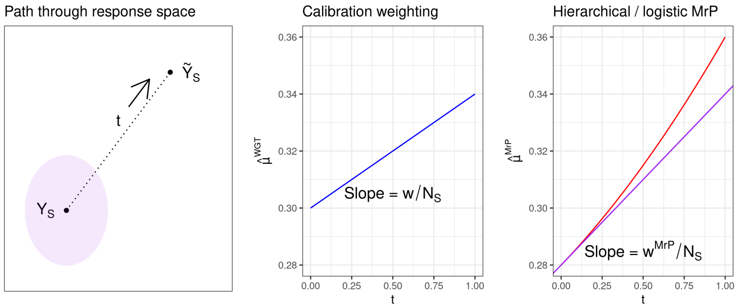

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$$w_i^{\mathrm{MrP}} := N_S \frac{\partial \hat{\mu}^{\mathrm{MrP}}(Y_{\mathcal{S}})}{\partial y_i} = N_S \mathrm{Cov}_{\mathcal{P}(\theta|Y_{\mathcal{S}})}\left(\frac{1}{N_T} \sum_{j=1}^{N_T} m(\theta^\intercal \mathbf{x}_j), \frac{\partial}{\partial y_i} \log \mathcal{P}(y_i|\mathbf{x}_i, \theta) \right)$$

Can estimate w_i^{MrP} *without re-running MCMC* [Gus96; GBJ18].

Same-Sex Marriage Survey Weights

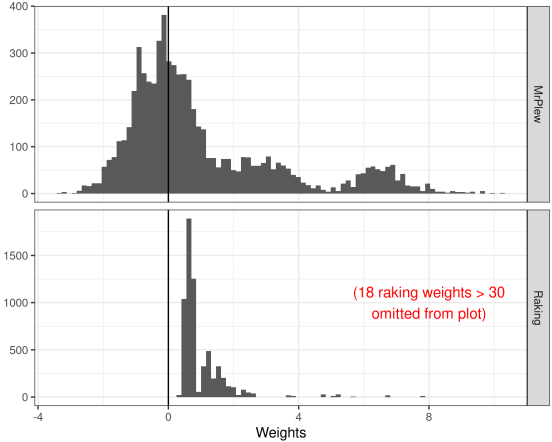

Figure 1: Comparison of weights for the Same-Sex Marriage analysis

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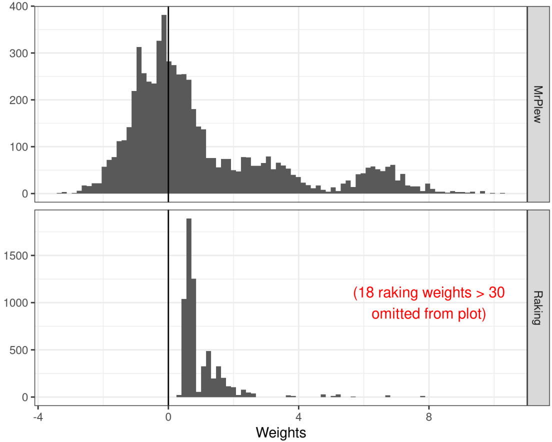

Figure 1: Comparison of weights for the Same-Sex Marriage analysis

What does the difference between these weights mean in terms of

- Frequentist sampling variance (standard errors)?
- Covariate balance / external validity?
- Subgroup contributions?

Same-Sex Marriage subgroup contribution

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Question:

How do responses in Texas affect our estimate of public opinion in CA?

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More specific question:

Suppose we *increased* the responses y_i in Texas.

How would our estimate of public opinion in CA change?

Take $\tilde{y}_i = y_i + \delta \mathbb{I}\left(i \text{ is in Texas}\right)$. Then $\hat{\mu}^{\mathrm{CW}}(\tilde{y}) - \hat{\mu}^{\mathrm{CW}}(y) = \delta \underbrace{\sum_{i \text{ in Texas}} w_i}_{\text{Texas's contribution}}$.

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Theorem: [Gio+26] (informal)

Take $\tilde{y}_i = y_i + \delta f(\mathbf{x}_i)$. We state conditions under which

$$\sup_{f \in \mathcal{F}} \left| \hat{\mu}^{\text{MrP}}(\tilde{Y}_{\mathcal{S}}) - \hat{\mu}^{\text{MrP}}(Y_{\mathcal{S}}) - \delta \sum_{i=1}^{N_{\mathcal{S}}} w_i^{\text{MrP}} f(\mathbf{x}_i) \right| = O(\delta^2) \text{ as } N \to \infty$$

...for a very broad class of $\mathcal{F}$. [3]

Uniformity helps justify searching over a large class of perturbations f.

[3] $\mathcal{F}$ can be any Donsker class of measurable functions with uniformly bounded $\mathbb{E}\left[||\mathbf{x}||\ f(\mathbf{x})\right]\ {}^2_2$.

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Summarize state contribution: $\sum_{i \text{ in state}} w_i$

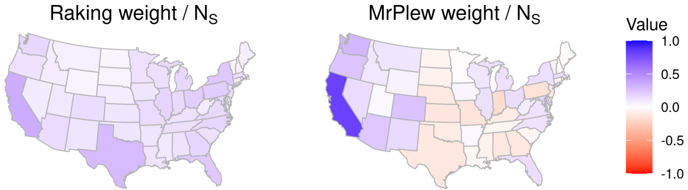

Figure 2: Subgroup contribution for Same-Sex Marriage in California

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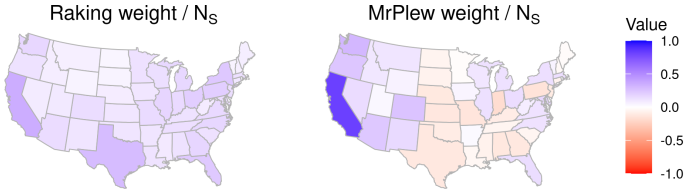

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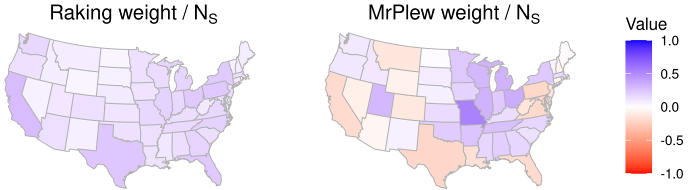

Figure 3: Subgroup contribution for Same-Sex Marriage in Missouri

Same-Sex Marriage frequentist variance

Theorem [Gio+26] (informal):

You can use the weights for frequentist variance as if MrP were linear if (and only if) y_i is a sufficient statistic of the model.

Frequentist variance of $\sqrt{N_S}\hat{\mu}^{\mathrm{MrP}}$ and $\sqrt{N_S}\hat{\mu}^{\mathrm{CW}}$
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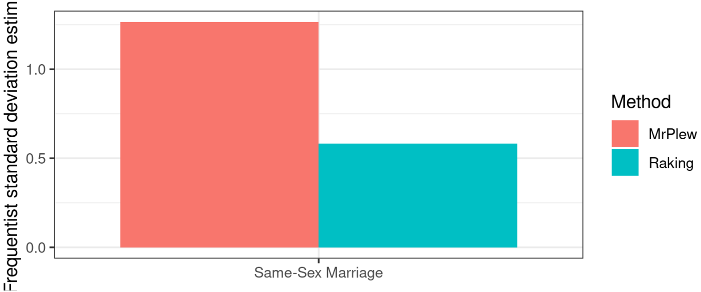

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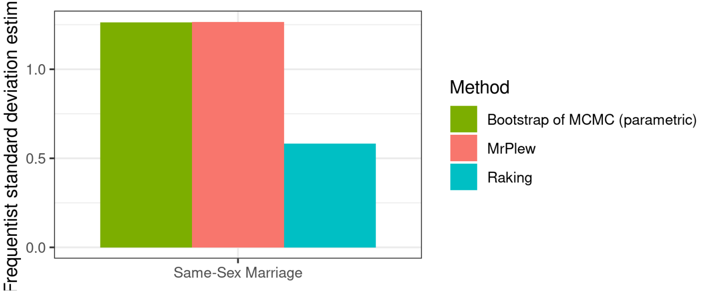

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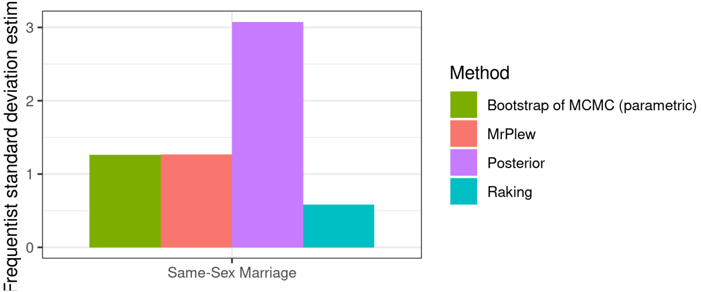
You can use the weights for frequentist variance as if MrP were linear if (and only if) y_i is a sufficient statistic of the model.

Frequentist variance of $\sqrt{N_S}\hat{\mu}^{\mathrm{MrP}}$ and $\sqrt{N_S}\hat{\mu}^{\mathrm{CW}}$
under re-sampling of survey data (keeping the target fixed)



Note that frequentist and posterior variance can be very different in misspecified models with many random effects!

Conclusion

By defining MrP locally equivlanent weights, we can rigorously provide a novel class of model checking metrics for non-linear MrP estimators.

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⚠ **Stop! That is a misleading conclusion!**

Conclusion

~~**By defining MrP locally equivlanent weights, we can rigorously provide a novel class of model checking metrics for non-linear MrP estimators.**~~

Stop! That is a misleading conclusion!

MrP is modeling the response, not the sampling bias.

CW-like checks are not diagnostics for the response model.

A correctly specified response model can do badly on CW checks, and vice-versa! (See paper.)

This work is best thought of as *model interrogation* for MrP.

Conclusion (let's try that again)

By defining *MrP locally equivlanent weights*, we can rigorously provide analogues of many CW diagnostics for non-linear MrP estimators for model interrogation.

(Similar ideas can to lead to actual checks of MrP model misspecification. Coming soon...)

Conclusion (let's try that again)

By defining *MrP locally equivlanent weights*, we can rigorously provide analogues of many CW diagnostics for non-linear MrP estimators for model interrogation.

(Similar ideas can to lead to actual checks of MrP model misspecification. Coming soon...)

R package available at
`https://github.com/rgiordan/mrplew`

Preprint at
`https://arxiv.org/abs/2606.04250`

Extra slides

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Same-Sex Marriage Survey Weights

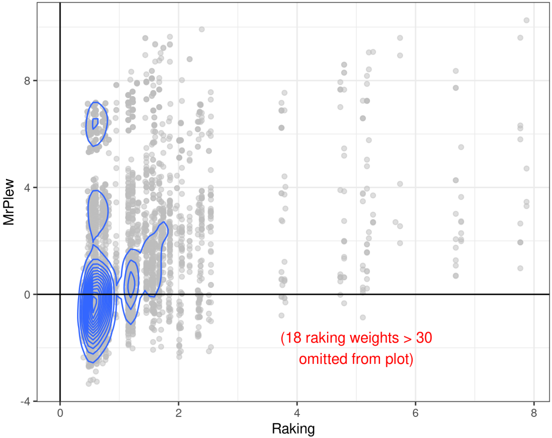

Figure 4: Comparison of weights for the Same-Sex Marriage analysis

Same-Sex Marriage covariate balance

Covariate balance / external consistency: $\frac{1}{N_T}\sum_{j=1}^{N_T} f(\mathbf{x}_j) \overset{\text{check}}{=} \frac{1}{N_S}\sum_{i=1}^{N_S} w_i f(\mathbf{x}_i)$